

Objective: To design a Multi-Function gate with 4 inputs to simulate the AND,NAND,OR,NOR logic gate operations using Vivado and implement it into the BASYS 3 circuit board to simulate its design's intention.

Equipment: Vivado 2017-4 software used to program a BASYS 3 circuit board.

Procedure/Design: The multi-function gate takes in 4 inputs:A,B,X, and Y, with 1 output: F. The X and Y input/switches represent which operation the A and B input switches are simulating. Since there are 4 inputs, there are $2^4 = 16$ different combinations. These combinations are represented by the truth table below:

X	Y	Operation of F
0	0	AND
0	1	OR
1	0	NOR
1	1	NAND

(X and Y)multipled by (A and B):

A	B
0	0
0	1
1	0
1	1

X	Y	A	B	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

The minterms are m(3,5,6,7,8,12,13,14) which simplify to $F = XA'B' + X'AB + YAB' + YA'B$ via Karnaugh map.

$$\bar{X}\bar{Y}AB + \bar{X}Y\bar{A}B + \bar{X}YAB\bar{B} + X\bar{Y}\bar{A}\bar{B} + X\bar{Y}AB$$

$$+ XY\bar{A}\bar{B} + XY\bar{A}B + XYAB$$

AB

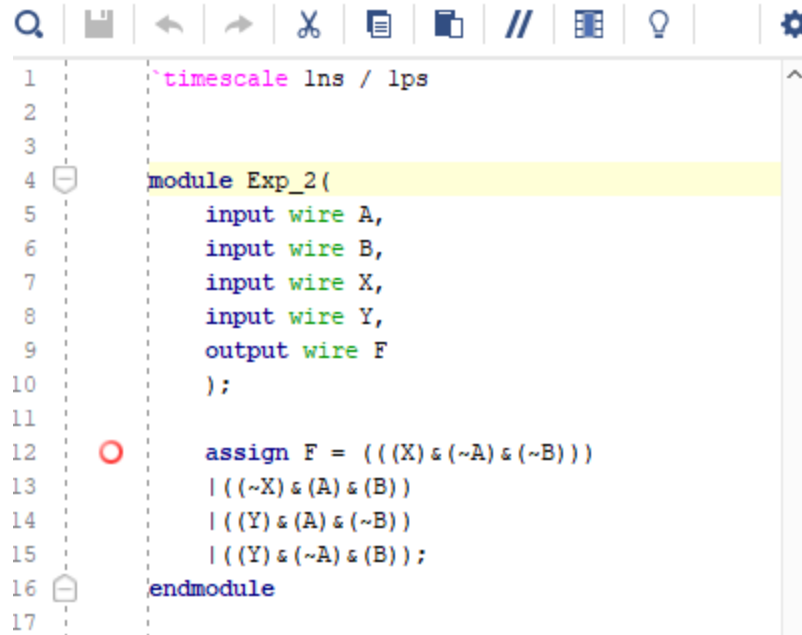
XY

	00	01	00	10
00	0	1	1	2
01	4	1	5	1
11	1	1	13	1
10	8	9	11	10

$$F \rightarrow X\bar{A}\bar{B} + \bar{X}AB + YAB + Y\bar{A}B$$

Source Code:

C:/Users/ry266542/Desktop/Lab/Exp_2.v



```
1  `timescale 1ns / 1ps
2
3
4  module Exp_2(
5      input wire A,
6      input wire B,
7      input wire X,
8      input wire Y,
9      output wire F
10 );
11
12  assign F = ((X) & (~A) & (~B))
13             | ((~X) & (A) & (B))
14             | ((Y) & (A) & (~B))
15             | ((Y) & (~A) & (B));
16 endmodule
17
```

Test Bench:

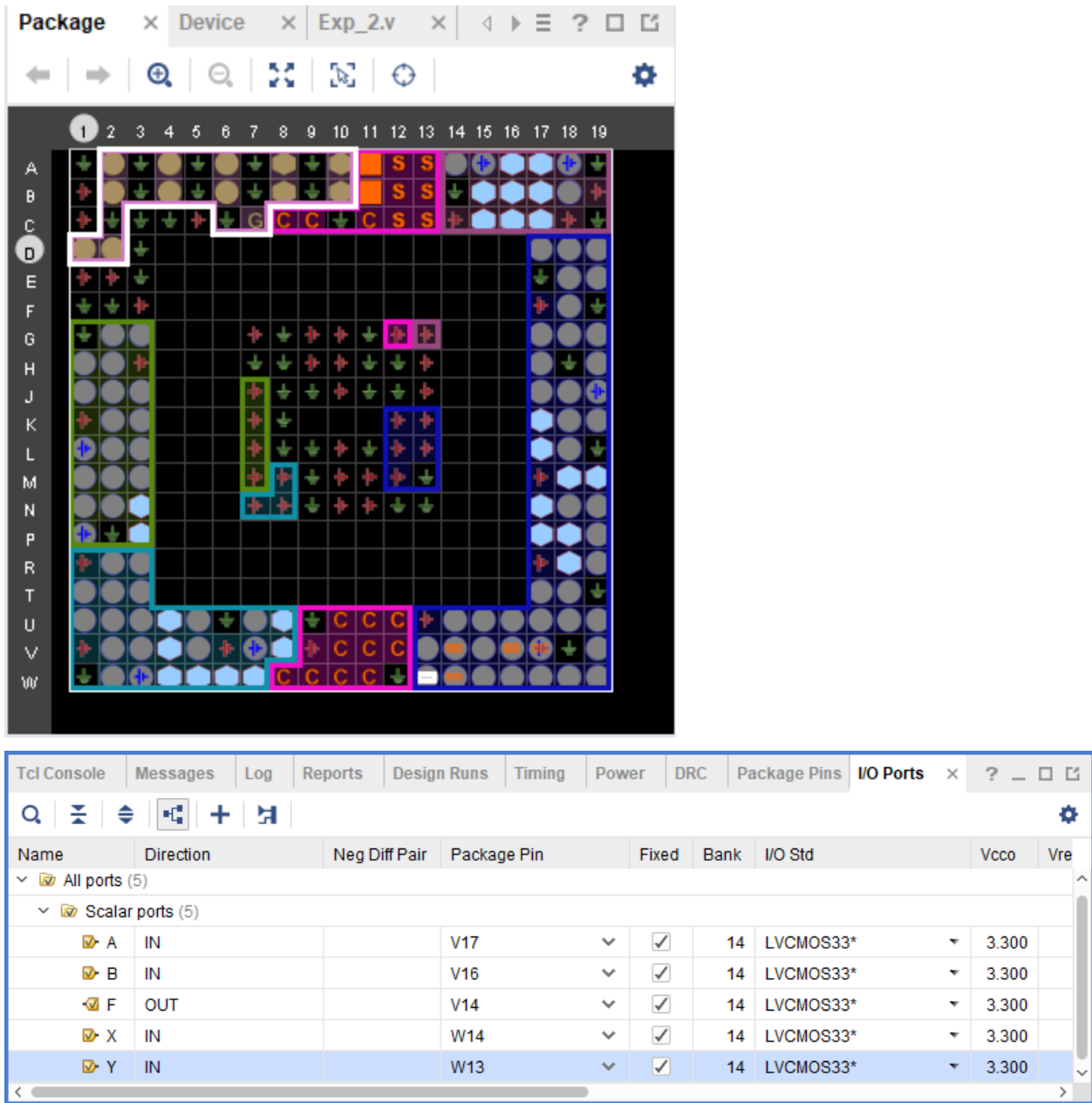
```

Exp2_Sim.v
C:/Users/ry266542/Desktop/Lab/Experiment 2/Experiment 2.srscs/sim_1/new/Exp2_Sim.v

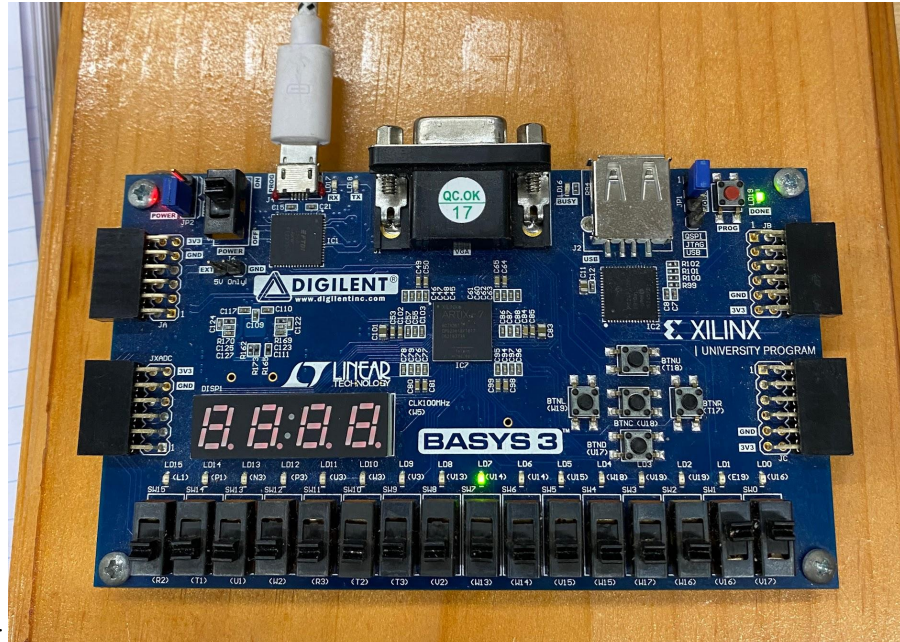
1  `timescale 1ns / 1ps
2
3  module Exp2_Sim(
4
5  );
6  //Inputs
7      reg A_t;
8      reg B_t;
9      reg X_t;
10     reg Y_t;
11     //Outputs
12     wire F_t;
13     //Instantiation of module or UUT(Unit Under Test)
14     Exp_2 UUT(
15         .A(A_t),
16         .B(B_t),
17         .X(X_t),
18         .Y(Y_t),
19         .F(F_t)
20     );
21     //Initialization of inputs
22     initial
23     begin
24         A_t=1'b0;
25         B_t=1'b0;
26         X_t=1'b0;
27         Y_t=1'b0;
28     end
29     //Test cases generation
30     always #10 A_t=~A_t;
31     always #20 B_t=~B_t;
32     always #30 X_t=~X_t;
33     always #40 Y_t=~Y_t;
34 endmodule
35

```

Assigned input/output:

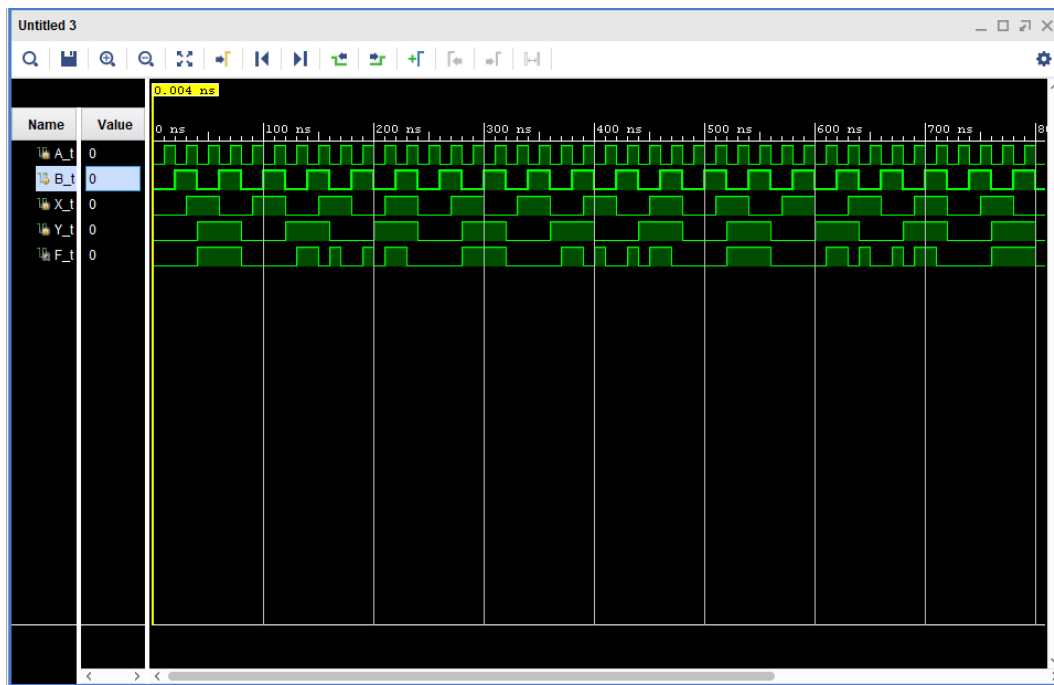


Results: The output of the experiment matched the intended truth table I shared above in the design.



$F=(X,Y,A,B): (0,0,1,1):$

This picture is from the On output of the AND operation.



Conclusion: The experiment was successful, the output from the BASYS 3 circuit board matched the output from the truth table using the corresponding input switches. This multi-function gate can be used by using the NAND and NOR operations since these operations are inverted versions of AND and OR. The number of outputs would change based on the

number of inputs: 2^n , n being the number of inputs, not affecting the operation lines. By changing the number of functions it will alter the operation lines, needing more or less lines as needed if we add or subtract functions from the experiment.